

HÉCTOR MANUEL ROMO-PARRA, Ph.D.

Neuroscientist · Clinical Psychologist · Cellular, Translational & Computational Neuroscience

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SCOPUS ID: 55334639500 | GitHub: [Hromo-parra/NeuroPsicoLocos](https://github.com/Hromo-parra/NeuroPsicoLocos) | SciProfiles: 4859859

PROFESSIONAL PROFILE

Neuroscientist and clinical psychologist with more than 20 years of experience in cellular and systems electrophysiology, experimental neuroscience, neurological disease models, neurodevelopmental disorders, quantitative neuroscience, and university teaching. Research interests span neuronal excitability, synaptic plasticity, epilepsy, cerebellar and hippocampal physiology, neuroinflammation and microglia, metabolic and mitochondrial mechanisms, cognition and memory, and computational phenotyping. Experienced in Python, MATLAB, machine learning, multivariate analysis, and statistical modeling applied to neuroscience and clinical data. International research training in Heidelberg and Münster, Germany, with additional clinical/psychiatric training. Author of 36 peer-reviewed publications listed in SciProfiles, including 9 publications in 2026, with sustained productivity across cellular, systems, neuroimmune, metabolic, and computational neuroscience.

RESEARCH METRICS & CURRENT STATUS

- 20+ years of neuroscience research experience.
- 36 publications listed in SciProfiles; h-index 18 as reported in current application materials.
- 9 publications in 2026, including work in Current Issues in Molecular Biology, Journal of Integrative Neuroscience, Brain Sciences, Epilepsy Research, Biomedicines, The Cerebellum, Neuroglia, and Ciencia y Salud.
- Member, Mexico's National System of Researchers (SNII), Level I; current institutional affiliation: National Institute of Neurology and Neurosurgery.
- Research Scientist, National Institute of Neurology and Neurosurgery, Mexico City.
- Full Professor, Universidad Iberoamericana, Mexico City.

RESEARCH INTERESTS

- Cellular and systems neurophysiology; in vivo and in vitro electrophysiology.
- Intrinsic neuronal excitability, ion channels, synaptic integration, long-term plasticity, and circuit state transitions.
- Experimental epilepsy and mechanisms linking seizures, plasticity, cognition, and memory.
- Cerebellar and hippocampal physiology.
- Neuroimmune mechanisms, microglial biology, neuroinflammation, and neurodegeneration.
- Metabolic and mitochondrial modulation of neuronal function; ketogenic and caloric-restriction interventions.
- Computational neuroscience, machine learning, multivariate phenotyping, and interpretable models.
- Neurodevelopmental and neuropsychiatric disorders; autism spectrum assessment.
- Human cognition, attention, memory, behavioral performance, and translational neuroscience.

PROFESSIONAL APPOINTMENTS

Research Scientist | National Institute of Neurology and Neurosurgery, Mexico City | 2022–Present

Research in neurophysiology and neurological disease, including epilepsy, neuronal excitability, neuroinflammation, metabolic mechanisms, and quantitative approaches to neuroscience.

Full Professor | Universidad Iberoamericana, Mexico City | 2022–Present

Teaching and mentorship in neuroscience, psychology, cognition, research methodology, and biologically grounded approaches to behavior; supervision of student research projects.

Research Professor & Neuroscience Area Coordinator | Universidad Anáhuac México | 2016–2022

Directed the Psychophysiology Laboratory; coordinated the Neuroscience area within the Faculty of Psychology; supervised graduate thesis research at M.Sc. and Ph.D. levels.

Senior Postdoctoral Researcher (Electrophysiology) | University of Münster, Germany | 2008–2015

Clinical Training in Psychiatry and Psychological Medicine | Universitätsklinikum Münster, Germany | 2008–2015

Postdoctoral Researcher (Electrophysiology) | University of Heidelberg, Germany | 2004–2008

EDUCATION & ADVANCED TRAINING

- Ph.D. in Sciences (Physiology / Neuroscience), UNAM & CINVESTAV, 2005.
- B.Sc. in Clinical Psychology, Summa Cum Laude, Universidad Salesiana (UNAM), 1998.

- International Certification: ADOS-2 and ADI-R.
- Postgraduate Training in Psychiatry, Vienna / Weill Cornell Medicine–Open Medical Institute, 2024.
- Diploma in Brief Therapy, CDHP, 2024.

SELECTED RECENT RESEARCH CONTRIBUTIONS

- Multivariate electrophysiological phenotyping of seizure-associated cerebellar Purkinje-neuron states, integrating cellular physiology with quantitative classification.
- Machine-learning approaches to epilepsy and multidimensional neurological data.
- Mitochondrial ion-channel and bioenergetic mechanisms relevant to antiseizure pharmacology.
- Metabolic mechanisms of epilepsy and the interaction between energy state and neuronal excitability.
- Microglial and neuroinflammatory mechanisms in neurological disease.
- Epilepsy, long-term potentiation, and implications for memory.
- Caloric restriction and Wnt/ β -catenin signaling in hippocampus and cortex in experimental epilepsy.
- Ketogenic interventions, neuroprotection, environmental neurotoxicity, and neurological disease.

RECENT PUBLICATIONS (2024–2026)

2026

- Meingüer-Cuevas MJ, Avila-García M, Espejel-Núñez A, Flores-Pliego A, Camacho-Arroyo I, et al. Emotional State and Salivary Inflammatory Markers in Endometriosis Associated Pelvic Pain: A Pilot Study Comparing Chronic and Cyclic Patterns. *Current Issues in Molecular Biology*. 2026;48(8):817.
- Garza-Álvarez AG, Romo-Parra H. Modelo estratégico para la atención médica resiliente en crisis sanitarias: análisis post-COVID-19. *Ciencia y Salud*. 2026.
- Rubio C, Romo-Parra H, López-Landa A, Rubio-Orsorio M. Correction: Rubio et al. Classification of Current Experimental Models of Epilepsy. *Brain Sciences*. 2026;16(4):400. doi:10.3390/brainsci16040400.
- Pérez-Villavicencio J, Martínez-Rojas VA, Rubio C, Serrano-García N, Galván EJ, Romo-Parra H. Machine learning in epilepsy. *Epilepsy Research*. 2026;223:107792. doi:10.1016/j.eplepsyres.2026.107792.
- Rubio C, Serrano-García N, Pérez-Rubio R, Pérez-Villavicencio J, Romo-Parra H, Lee Á, Rubio-Orsorio M. Antiseizure Medications Impact Mitochondrial Ion Channels via Novel Bioenergetic and Neural Mechanisms. *Journal of Integrative Neuroscience*. 2026;25(6):52316. doi:10.31083/JIN52316.
- Rubio C, Carpinteyro S, Serrano-García N, Romo-Parra H, Pérez-Villavicencio J, Lee Á, Mercado-Pimentel R, Rubio-Orsorio M. The Hidden Metabolic Roots of Epilepsy. *Biomedicines*. 2026;14(4):764. doi:10.3390/biomedicines14040764.
- Ganado M, Rubio C, Pérez-Villavicencio J, Serrano N, Romo-Parra H, Lee Á. Opioid Receptors in Psychedelia: Indirect Serotonergic Modulation of Direct KOR Activation by Salvinorin A. *Biomedicines*. 2026;14(2):476.
- Moreno W, Martínez-Rojas VA, Galván EJ, Sierra-Ramírez JA, Rubio-Orsorio M, Romo-Parra H, Rubio C. Seizures Reprogram Cerebellar Purkinje Neurons: A Multivariate Electrophysiological Classification Reveals Hidden Subtypes. *Cerebellum*. 2026;25(1):6. doi:10.1007/s12311-025-01949-1.
- Rubio-Orsorio M, Rubio C, Ganado M, Romo-Parra H. The Double Face of Microglia in the Brain. *Neuroglia*. 2026;7(1):3. doi:10.3390/neuroglia7010003.

2025

- Rubio C, López-Landa A, Serrano-García N, Romo-Parra H, Rubio-Orsorio M. Impact of Particulate Matter 2.5 on Neurological Diseases: Insights Into Pathophysiological and Molecular Mechanisms. *Journal of Toxicology*. 2025;5752904.
- Rubio C, Vázquez D, Pacheco Aispuro G, Romo H, Rubio-Orsorio M. Anti-neuroinflammatory therapies in epilepsy: A bibliometric analysis. *International Journal of Clinical Studies & Medical Case Reports*. 2025;50(5):001247.
- Marín-Castañeda LA, Pacheco Aispuro G, González-Garibay G, Martínez Zamora CA, Romo-Parra H, Rubio-Orsorio M, Rubio C. Interplay of epilepsy and long-term potentiation: implications for memory. *Frontiers in Neuroscience*. 2025;18:1451740.
- Uribe E, López-Landa A, Serrano-García N, Ochoa E, Marín-Castañeda LA, Vázquez D, Orozco S, Romo-Parra H, González-Garibay G, Rubio-Orsorio M, Rubio C. Impact of caloric restriction on the Wnt- β -catenin pathway in the hippocampus and cortex of a kindled rat model. *European Journal of Clinical and Experimental Medicine*. 2025;23(1):6–14.
- Rubio C, López-Landa A, Romo-Parra H, Rubio-Orsorio M. Impact of the Ketogenic Diet on Neurological Diseases: A Review. *Life*. 2025;15(1):71.

2024

- Rodríguez P, López-Landa A, Romo-Parra H, Rubio-Orsorio M, Rubio C. Unraveling the Ozone Impact and Oxidative Stress on the Nervous System. *Toxicology*. 2024;15(3):973.
- Rubio C, Romo-Parra H, López-Landa A, Rubio-Orsorio M. Classification of Current Experimental Models of Epilepsy. *Brain Sciences*. 2024;14(10):1024.

- Hernández-Suárez Á, Marín-Castañeda LA, Rubio C, Romo-Parra H. Effect of cannabidiol as a neuroprotective agent on neurodevelopmental impairment in rats with neonatal hypoxia. *Brain & Development*. 2024;46(9):294–301.
- Ochoa E, Gatica F, Morales Z, Romo-Parra H, Marín-Castañeda LA, Rubio C. The Ketogenic Diet and Alzheimer's Disease. In: *The Ketogenic Diet Reexamined: Myth vs. Reality*. NOVA Science Publishers. 2024.

Complete bibliographic record: 36 publications listed in SciProfiles; selected recent publications are shown above.

TEACHING, MENTORSHIP & ACADEMIC LEADERSHIP

- Undergraduate and graduate teaching in neuroscience, neurobiology, cognition, psychology, research methodology, and related biomedical topics.
- Supervision and mentoring of experimental and computational neuroscience projects, including graduate thesis work.
- Former Neuroscience Area Coordinator and Director of the Psychophysiology Laboratory, Universidad Anáhuac México.
- Teaching philosophy emphasizes mechanistic reasoning, quantitative literacy, reproducibility, experimental design, and critical use of AI-assisted research tools.
- Invited continuing-education teaching, including *Neurobiology of Addictions* (2024).

EDITORIAL, REVIEW & SCIENTIFIC SERVICE

- Guest/Contributing Editor, *Neuroglia* Special Issue focused on glial and neuroinflammatory mechanisms in neurological/neurodevelopmental disorders.
- Editorial role, *Ciencia y Salud*.
- Peer Reviewer: *Frontiers*, MDPI journals, Elsevier journals, *Epilepsia*, and other scientific journals.
- Accredited Evaluator, RCEA-SINECYT / CONACYT, since 2013.
- Evaluator, PNPC-CONACYT.

PROFESSIONAL AFFILIATIONS

- Society for Neuroscience (USA).
- American Psychological Association.
- Neuropsychanalysis Association.
- Open Medical Institute Fellow.

TECHNICAL & QUANTITATIVE SKILLS

- Electrophysiology: in vivo / in vitro; cellular excitability, synaptic physiology, pClamp Suite, Patchmaster Suite.
- Programming and analysis: Python, MATLAB, C; supervised and unsupervised machine learning; multivariate statistics; dimensionality reduction; statistical modeling.
- Data and visualization: OriginLab, GraphPad; reproducible quantitative workflows.
- Clinical/diagnostic: ADOS-2 and ADI-R, internationally certified.
- AI-assisted research and large language model tools for scientific workflows, coding, literature synthesis, and education.

LANGUAGES

- Spanish: native.
- English: advanced.
- German: advanced.

ACADEMIC SCHOLARSHIPS & SPONSORSHIP

- UNAM undergraduate scholarship, 1994–1997.
- Boehringer Ingelheim sponsorship supporting undergraduate thesis work, 1997–1999.
- UNAM merit-based doctoral scholarship awarded for academic performance (10/10 record).
- CONACYT doctoral scholarship during Ph.D. training, 2000–2005. Completed; no ongoing financial obligation or commitment associated with the award.

SCIENCE COMMUNICATION & DIGITAL SCHOLARSHIP

- Founder and Editor, *Neuropsicolocos*, a Spanish-language neuroscience communication and education platform.
- Development of interactive neuroscience and psychology teaching resources, experimental web applications, computational demonstrations, and open educational tools.
- GitHub-based development of neuroscience teaching, experimental, and quantitative projects.